GSEA

GSEA for proteomics data

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The script performs GSEA to identify biological pathways enriched in the differential expression results of proteomics data from Cancer Cell Lines.

Workflow overview:

1. Downloading Hallmark GSEA results from MSigDB (GSEA)
2. Gene Set Prep: Loads pathways from multiple sources (GO/KEGG)
3. Ranking: Ranks genes by log10 fdr (sign(logFC) for every comparison
4. Analysis Loop: Runs GSEA for every combination of Comparison x Gene Set Database)
5. Visualizations:
   - Generates summary bar plots of top normalized enrichment scores (NES)
   - Generates specific enrichment plots (enrichment plots) for top pathways

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Set directories

```
# Clear
rm(list=ls())

# Load libraries
library(dplyr)
library(ggplot2)

gsf_dir = "/home/jupyter-jovyan-lisa/gsea_results"
dir.create(file.path(gsf_dir), showWarnings = FALSE)
dir.create(file.path(gsf_dir,"plots"), showWarnings = FALSE)
dir.create(file.path(gsf_dir,"plots/summary_plots"), showWarnings = FALSE)
dir.create(file.path(gsf_dir,"plots/enrichment_plots"), showWarnings = FALSE)
```

Load data

```
# Load data
data_dir = "/home/jupyter-jovyan-lisa/DE/Cancer_Cell_Lines/Proteomics/Results/"
de_data = read.csv(paste0(data_dir,"Proteomics_DE_Cancer_Cell_Lines.csv"))
de_data = de_data[de_data$Comparison == "Proteomics",]
de_data = de_data[de_data$Contrast == "Tumor_vs_Normal",]
de_data = de_data[de_data$Tissue == "Lung",]
de_data = de_data[de_data$Model == "Tumor",]
de_data = de_data[de_data$Treatment == "Control",]
de_data = de_data[de_data$Time == "24",]
de_data = de_data[de_data$Dose == "0",]
de_data = de_data[de_data$Cell_Line == "A549",]
de_data = de_data[de_data$Gene != "",]
de_data = de_data[de_data$Gene != NA,]
de_data = de_data[de_data$adj.P.Val < 0.05,]
de_data = de_data[de_data$logFC > 0,]
de_data = de_data[de_data$Gene != "Q9Y6Q6",]
de_data = de_data[de_data$Gene != "P08670",]
de_data = de_data[de_data$Gene != "P02533",]
de_data = de_data[de_data$Gene != "P02538",]
de_data = de_data[de_data$Gene != "P05787",]
de_data = de_data[de_data$Gene != "P08729",]
de_data = de_data[de_data$Gene != "P13645",]
de_data = de_data[de_data$Gene != "P35527",]
de_data = de_data[de_data$Gene != "P02768",]
de_data = de_data[de_data$Gene != "P02787",]
de_data = de_data[de_data$Gene != "P02788",]
de_data = de_data[de_data$Gene != "P02790",]
de_data = de_data[de_data$Gene != "P02792",]
de_data = de_data[de_data$Gene != "P02794",]
de_data = de_data[de_data$Gene != "P02795",]
de_data = de_data[de_data$Gene != "P02796",]
de_data = de_data[de_data$Gene != "P02797",]
de_data = de_data[de_data$Gene != "P02798",]
de_data = de_data[de_data$Gene != "P02799",]
de_data = de_data[de_data$Gene != "P02800",]
de_data = de_data[de_data$Gene != "P02801",]
de_data = de_data[de_data$Gene !=
```